Final Project - Bayesian Analysis of Alzheimers Data

Problem Formulation:

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia, characterized by declining memory and cognition. Early detection through biomarkers and MRI data is critical, as delayed treatment is associated with worse outcomes. In this project, we investigate clinical risk factors associated with Alzheimer's disease using Bayesian Statistics, aiming to identify which groups of people are at higher risk and support early intervention.

Our research questions are:

i) **How are brain volume, age and biological sex associated with the probability of receiving a diagnosis of Alzheimer's disease and the diagnosed CDR (Clinical dementia rating)**

- Is there an effect of sex on Alzheimer's disease probability?
- Do individuals with lower brain volume have higher CDR scores?
- Does higher education levels dampen the effect of brain volume on CDR?

Basic Biological Context and Important Variables

CDR (Clinical Dementia Rating) is a common tool used to assess cognitive impairment, with scores mapped as follows:
0: none 0.5: questionable 1: mild 2: moderate 3: severe
Any score > 0 indicates probable AD, making CDR both a diagnostic and severity measure.

MMSE: (Mini-Mental State Examination score) This is a brief test of global cognitive function, with the following scores describing the severity of cognitive decline.
24–30: Normal 19–23: Mild 10–18: Moderate 0–9: Severe

eTIV (estimated total intracranial volume) Estimated total volume inside skull, estimated from MRI results.

nWBV (normalized whole-brain volume) This is the proportion of the intracranial cavity occupied by brain tissue. A lower volume would indicate more brain atrophy. Normalization ensures that differences in nWBV correspond to brain atrophy instead of differences in head size. We are interested in this variable because brain atrophy is a key characteristic of Alzheimer's disease progression.

Educ (Education Level) describes an individual's highest level of education:
1: <HS 2: HS grad 3: some college 4: bachelor's 5: postgraduate

Key Modelling Challenges:

The key modelling challenge is that the response variable, CDR, is an ordinal variable, with higher values indicating higher disease severity. An ideal Bayesian model using our dataset has to take into account the structure of the categories. We will begin with naive models treating CDR as a binary random variable or as an unordered categorical variable. Then we would build upon these with a more sophisticated model to include the ordering. This is grounded in the framework of progressively increasing model complexity, as often done in the course.

In addition, we will assess the posterior predictive probabilities for our model through cross-validation, a tool that we were introduced to in the course. We will apply concepts from the theme of Bayesian calibration in this section.

Literature Review

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cognitive decline and structural brain atrophy. Neuroimaging studies have consistently shown that reductions in brain volume are strongly associated with both aging and Alzheimer's disease severity. In particular, whole brain volume is known to decline gradually with age and more rapidly in individuals with Alzheimer's disease, reflecting disease-related neurodegeneration. These structural changes are closely linked to clinical measures such as the CDR and MMSE.

A widely used dataset for studying these relationships is the Open Access Series of Imaging Studies (OASIS), introduced by Marcus et al. (2007). This dataset combines MRI-derived brain measurements with demographic and clinical variables, making it particularly suitable for statistical modeling of Alzheimer's disease. Key variables include age, biological sex (M/F), Educ, as well as brain imaging measures such as eTIV and nWBV.

Among these, nWBV is of particular importance as it provides a normalized measure of brain volume that captures atrophy, while eTIV reflects head size and serves as a scaling factor. Clinical variables such as CDR and MMSE provide complementary measures of disease presence and severity. Socioeconomic status (SES) and education level are also important covariates, as they capture social and cognitive factors that may influence disease risk and progression.

Previous studies using the OASIS dataset and similar neuroimaging data have primarily focused on identifying associations between brain structure and cognitive decline. For example, Fotenos et al. (2005) and Buckner et al. (2004) demonstrated strong relationships between brain atrophy and Alzheimer's disease severity. In addition, a large body of work has applied machine learning techniques to predict Alzheimer's disease from MRI data. Methods such as support vector machines (Klöppel et al., 2008) and deep learning models (Suk et al., 2014) achieve high classification accuracy by leveraging imaging features such as brain volume. However, these approaches typically emphasize predictive performance and often provide limited interpretability or uncertainty quantification.

In contrast, Bayesian statistical methods offer a principled framework for modeling uncertainty and incorporating prior knowledge in the analysis of complex biomedical data. Rather than producing point estimates, Bayesian approaches yield full posterior distributions for model parameters, allowing researchers to quantify uncertainty through credible intervals and probabilistic statements. This is particularly valuable in medical contexts, where uncertainty plays a critical role in decision-making.

Bayesian methods have been successfully applied in Alzheimer's research to model disease progression and cognitive decline. For instance, Fonteijn et al. (2012) introduced a Bayesian event-based model to characterize the progression of Alzheimer's disease as a sequence of latent events, while Donohue et al. (2014) used Bayesian hierarchical models to analyze longitudinal cognitive data (different datasets). These approaches demonstrate the flexibility of Bayesian modeling in capturing complex relationships and latent structures in neurodegenerative diseases.

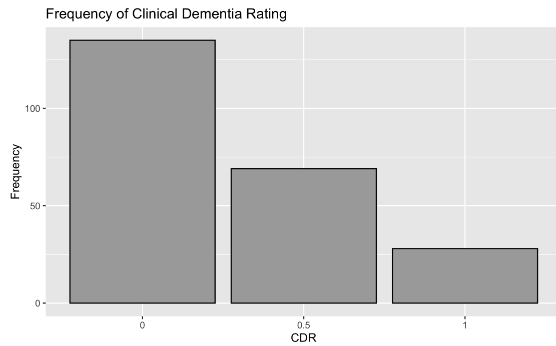
Despite these advantages, Bayesian approaches have been relatively underutilized in studies based on the OASIS dataset, where most analyses rely on frequentist regression or machine learning techniques.

Summary

Existing literature establishes strong links between brain atrophy, demographic factors, and Alzheimer's disease. However, it largely relies on predictive or frequentist approaches. By adopting a Bayesian perspective, this study contributes to the literature by providing interpretable, uncertainty-aware estimates of how brain structure and clinical variables jointly influence Alzheimer's disease risk and severity.

Exploratory Data Analysis

First, we must explore the dataset's feature distributions before designing and fitting models.

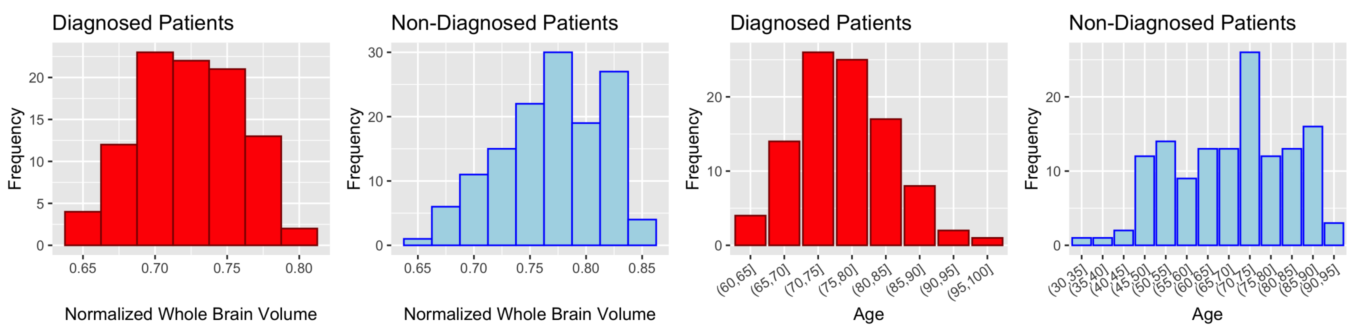


Most patients are not diagnosed with Alzheimers, and of the diagnosed patients, most of them have been diagnosed with only mild dementia.

Table 1: Diagnosis vs Sex

Biological_Sex	Diagnosed	NotDiagnosed
Female	57	97
Male	40	38

There are more females in the dataset, males have a higher diagnosis rate than females.



From the plot above, it appears diagnosed patients typically have less whole brain volume than non-diagnosed patients. Diagnosed patients tend to be older, where non diagnosed patients represent all ages.

Models

Naive Bayesian Logistic Regression Model

Priors

$$\beta_j \sim \mathcal{N}(0, 3), \quad \text{for } j \in \{0, 1, 2, 3, 4, 43\}.$$

Linear Predictor

$$\begin{aligned}\eta_i = \beta_0 + \beta_1 \cdot \text{sex}_{i,\text{male}} + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{brain\_vol}_i + \beta_4 \cdot \text{educ}_i \\ + \beta_{34} \cdot (\text{brain\_vol}_i \times \text{educ}_i).\end{aligned}$$

Link Function

$$\theta_i = logistic(\eta_i)$$

Likelihood

$$\text{CDR\_binary}_i \mid \theta_i \sim \text{Bernoulli}(\theta_i).$$

Complex Model

Note that we have set $\beta_0 = 1$ for identifiability

Priors

$$\begin{aligned}
\beta_j &\sim \mathcal{N}(0, 2) \quad (j = 1, 2, 3, 34)\\
\delta &\sim \text{Exp}\,(0.3)\\
k_0 &\sim \mathcal{N}(0, 1), \quad k_1 = k_0 + \delta\\
\sigma &\sim \text{Exponential}(0.3)
\end{aligned}$$

Likelihood

Hidden Disease Severity Score

$$\begin{aligned}
\mu(i) = \beta_0 &+ \beta_1\, \text{sex_male}_i + \beta_2\, \text{age}_i + \beta_3\, \text{brain_vol}_i + \beta_4\, \text{educ}_i\\
&+ \beta_{3,4}\, (\text{brain_vol}_i \times \text{educ}_i)
\end{aligned}$$

$$Z_i \mid \mu(i) \sim \mathcal{N}(\mu(i), \sigma)$$

Disease Outcome

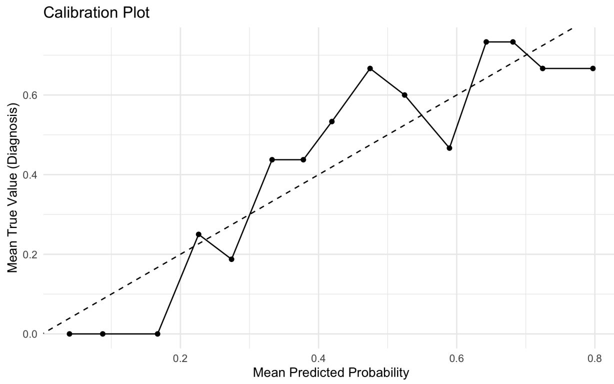
$$Y_i \mid Z_i = \begin{cases} 0 & \text{if } Z_i \leq k_0 \\ 0.5 & \text{if } k_0 \leq Z_i \leq k_1 \\ 1 & \text{if } Z_i > k_1 \end{cases}$$

Implementation of the Naive Model

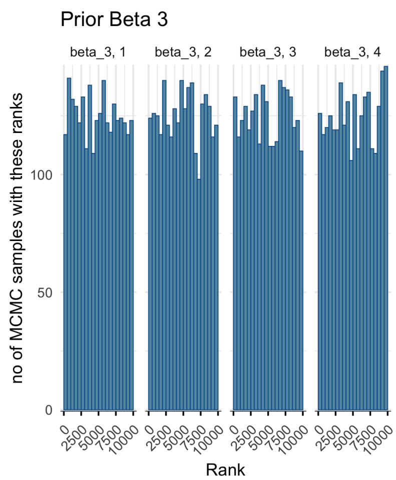
After fitting the model, we inspect the quality of the fit.

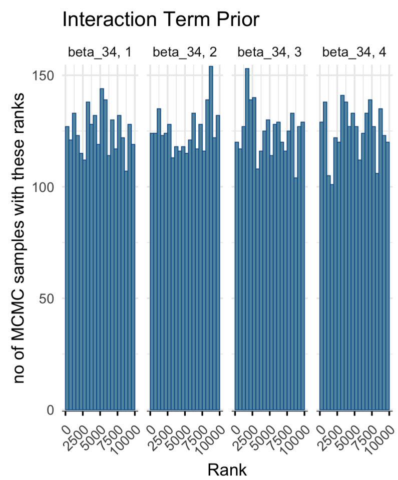
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp___	-130.3020475	-129.9521250	1.7434656	1.5959077	-133.6149625	-128.115848	0.9999957	4014.000	5753.700
beta_0	1.3735945	1.3821760	1.9814129	1.9713963	-1.8674770	4.617477	1.0009348	4671.063	5136.355
beta_1	0.6363919	0.6345535	0.3282910	0.3308136	0.1028844	1.182588	1.0002459	6114.409	5174.584
beta_2	0.0144233	0.0140244	0.0163888	0.0162147	-0.0122494	0.042013	1.0002739	5109.413	5420.891
beta_3	-2.3942652	-2.3745006	2.2841749	2.2473440	-6.1081557	1.375524	1.0006261	4364.827	5052.990
beta_4	3.2911694	3.2811631	1.0223509	1.0184403	1.6269091	4.988054	1.0003153	3802.224	4736.381
beta_34	-4.9263032	-4.9299621	1.3879566	1.3829218	-7.2362108	-2.665211	1.0002284	3802.930	4881.806

`rhat` ≈ 1 for all parameters, indicating that the chains have all converged to the same distribution. The bulk and tail ESS are high, indicating efficient sampling.

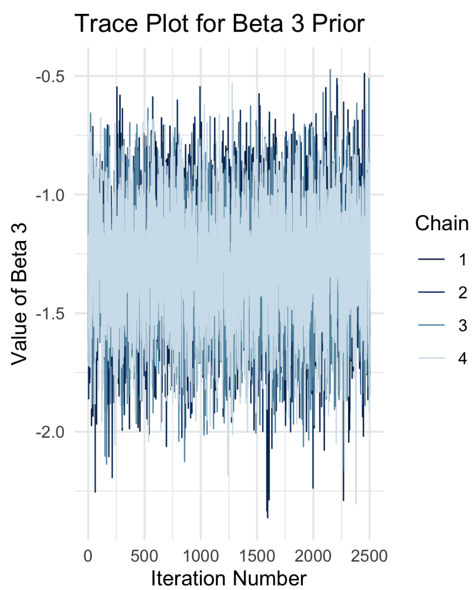


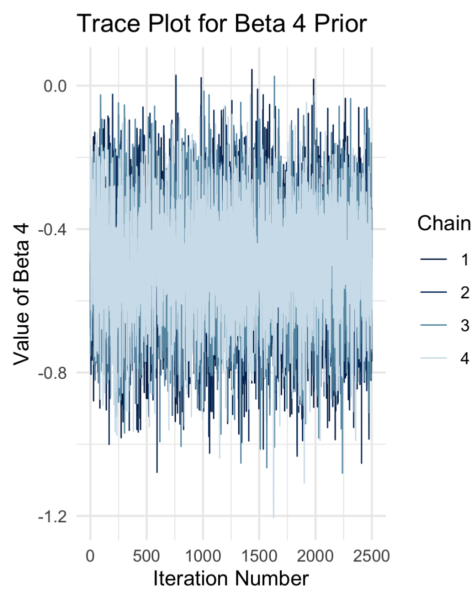
As we can see, the model is only decently calibrated.





All of the histograms look roughly flat for all 4 chains for the parameters displayed. This is an indication of good mixing.





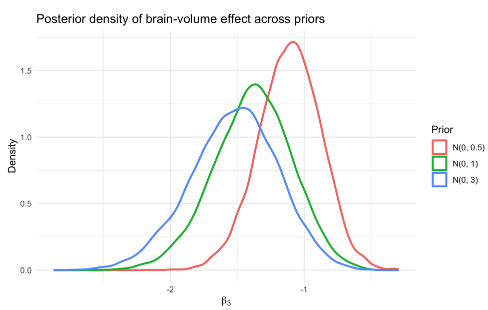
All 4 chains have plenty of overlap, and don't appear to be stuck at any point. This demonstrates good mixing for all of the priors shown.

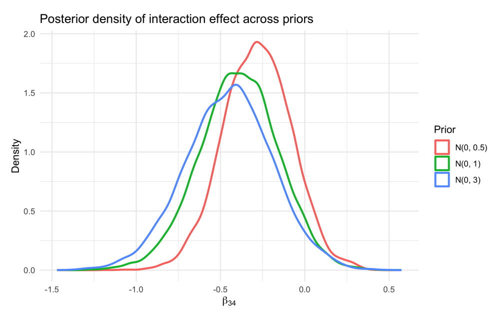
```
##
## Computed from 10000 by 232 log-likelihood matrix.
##
##          Estimate   SE
## elpd_loo   -129.1  7.6
## p_loo         6.1  0.7
## looic       258.2 15.1
## ------
## MCSE of elpd_loo is 0.0.
## MCSE and ESS estimates assume independent draws (r_eff=1).
##
## All Pareto k estimates are good (k < 0.7).
## See help('pareto-k-diagnostic') for details.
```

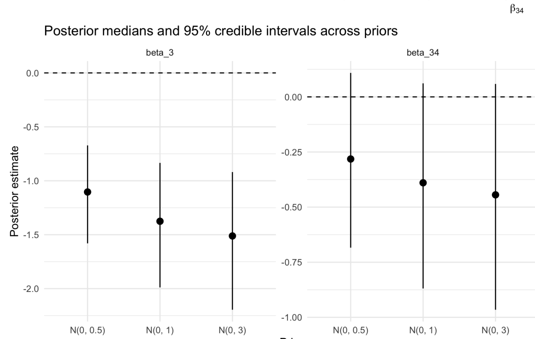
```
## Average probability assigned to true outcome: 0.573
```

The Pareto k estimates tell us that there are no outliers influencing the model.
The model is assigning 57.3% probability to the correct outcome (correct class of dementia diagnosis) on average for the leave-one-out predictions.

Sensitivity Analysis for Naive Model





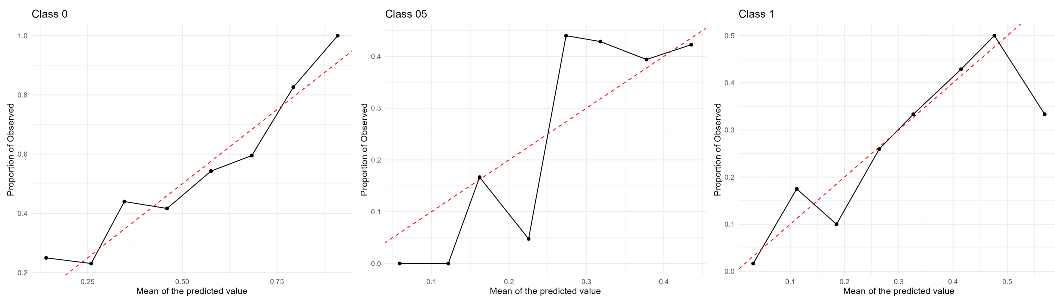


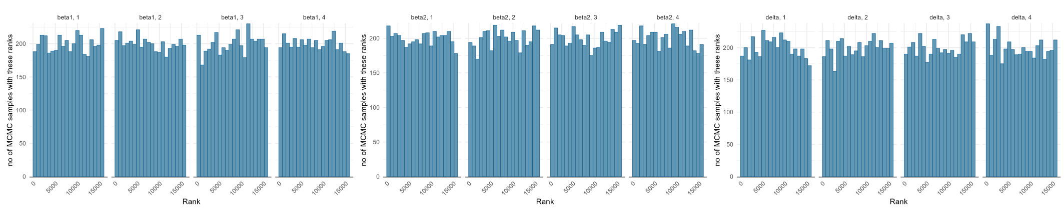
Implementation of the Complex Model

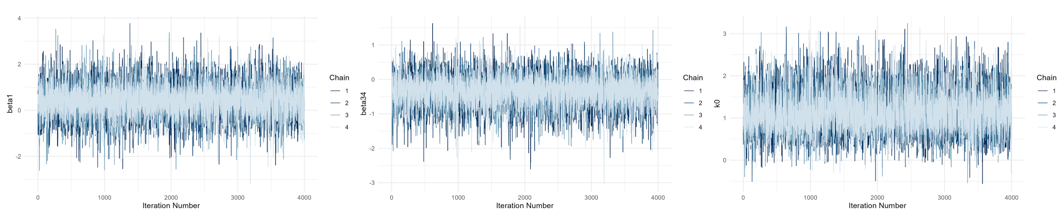
After fitting, we inspect the quality of the fit and calibration of the model.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-182.6412980	-182.2613850	2.5612897	2.4478467	-187.309628	-179.1594335	1.001004	3345.428	6760.190
beta1	0.3948292	0.3863903	0.7147512	0.6538026	-0.776991	1.5766076	1.000827	10169.697	8113.184
beta2	-0.2910655	-0.2953637	0.5587883	0.5098471	-1.188606	0.6165874	1.000562	10369.118	9198.305

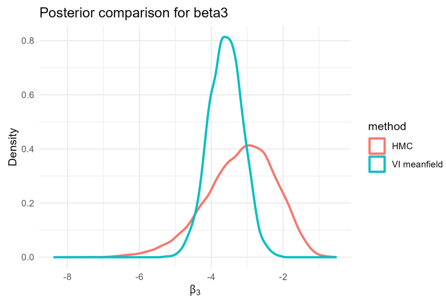
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
beta3	-3.1895708	-3.1107940	0.9943077	0.9861639	-4.958251	-1.6991747	1.001118	3361.062	3856.037
beta4	-1.1485452	-1.0823258	0.5125464	0.4818367	-2.082053	-0.4281073	1.000494	5472.295	8342.659
beta34	-0.3443083	-0.3235710	0.4307925	0.3909391	-1.083524	0.3260296	1.000417	10381.270	8908.580

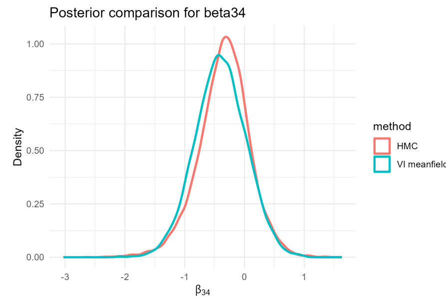


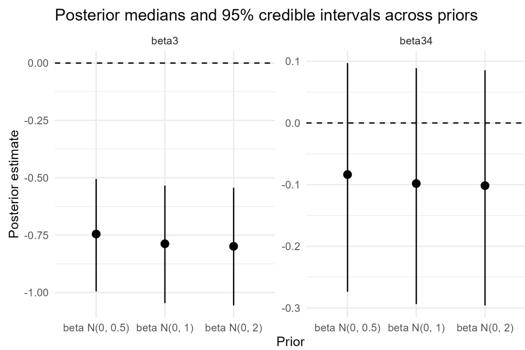




Senstivity Analysis on the Complex Model







Results and Discussion

We will now try to tackle some of the questions we were interested in using the ordinal model we introduced. From our exploratory analysis, it seemed that males had a greater chance of being diagnosed. We now try to frame this in terms of the query "Is the coefficient of sex_male in our model greater than 0?. We try to answer this in two ways:

1. *Posterior Probability*: We estimated

- $P(\beta_1 > 0|Data) = 0.726$: The posterior probability of $\beta_1 > 0$ in the complex model was 0.726.
- $\mathbb{E}(\beta_1|Data) = 0.395$ and a 90% credible interval is $[-0.777, 1.577]$.

From these results, it seems that we have some evidence that keeping the other variables constant, biological males usually have a higher CDR score than females. We now approach this question from a decision theory standpoint.

2. *Decision Theory Framework*: We wish to make a decision whether $\beta_1 > 0$. Let the action $a = 1$ mean concluding that there is a positive slope β_1. Define the loss function:

$$L(a, \beta_1) = 2\mathbb{I}(a = 1, \beta_1 < 0) + \mathbb{I}(a = 0, \beta > 0)$$

where $a = 1$ indicates that we claim the slope is > 0. This loss function was chosen because we would like to penalise a false positive much more than a false negative. Using this loss function we estimate

$L(a) := \mathbb{E}(L(a, X)) = 0.548\mathbb{I}(a = 1) + 0.726\mathbb{I}(a = 0)$.

Using this framework, under this loss function, an optimal decision would be to conclude that the sex_male effect is positive.

Now, similarly to answer our second and third questions we pose the query: What is the probability of $\beta_3 + \beta_{34} educ_i < 0$? Using three different level of standardized education levels $(-1, 0, 1)$, we found that for each of the three levels, the posterior probability $Pr(\beta_3 + \beta_{34} * educ_i < 0|Data) \approx 1$. This table summarises the posterior means and credible intervals

Education Level	Posterior Mean of Brain Volume Effect	95% CrI	P(effect < 0)
Low (-1)	-2.845	(-5.015, -1.229)	1.00
Mean (0)	-3.190	(-5.355, -1.491)	1.00
High (-1)	-3.534	(-6.13, -1.565)	1.00

This gives strong evidence that that individuals with lower brain volume have higher CDR scores controlling for the other variables. Since the evidence is so overwhelming, we do not treat this with a decision theoretic framework.

To answer our final question, we need to ask whether the interaction term in our complex model β_{34} is greater than 0. A positive term would mean that at higher education levels the effect of reducing brain volume on the severity of the disease is smaller.

1. *Posterior Probability*: We found

- $Pr(\beta_{34} > 0|Data) \approx 0.1948125$, which is very small and incdicates we are not very certain that this coefficient is positive.

2. *Decision Theory Framework*: Using the same loss function and actions as before, we calculated $L(a) := \mathbb{E}(L(a, X)) = 1.610375\mathbb{I}(a = 1) + 0.1948\mathbb{I}(a = 0)$, so the Bayes estimator would be to conclude that there isn't a dampening effect of education.

Prior Sensitivity Analysis

We assessed robustness to prior choice by varying the prior variance for the regression coefficients.

For the complex model, the posterior for the brain-volume effect β_3 was highly stable across priors. The posterior median changed only slightly, and the credible intervals remained clearly below zero, indicating that this effect is strongly informed by the data. The interaction term β_{34} was also relatively stable, though its credible interval remained wider and closer to zero, suggesting weaker evidence.

In contrast, the naive model showed greater sensitivity to prior choice. Both β_3 and β_{34} shifted more noticeably as the prior variance increased, and their credible intervals widened. This indicates weaker identifiability and a stronger influence of the prior.

Overall, the complex model provides more robust, data-driven inference, while the naive model is more sensitive to prior assumptions.

Comparison of HMC and Variational Inference

We compared HMC with mean-field variational inference (VI) for the complex model.

For β_3, VI produced a noticeably narrower posterior than HMC, indicating underestimation of uncertainty. This is consistent with the limitations of mean-field VI, which cannot capture posterior dependence between parameters. Despite this, both methods agreed on the negative direction of the effect.

For β_{34}, the HMC and VI posteriors were much more similar in both location and spread, suggesting that this parameter is easier to approximate with a factorized variational family.

Since HMC diagnostics indicated good convergence, the broader HMC posterior should be considered more reliable. Thus, while VI recovers posterior means well, HMC provides more accurate uncertainty quantification.

Limitations and Implication of Results

Our analysis has a few limitations. Firstly, we did not have a lot of data on the classes 0.5 and 1 and no data for diagnosis levels above that. Since CDR is an ordinal scale ranging from 0 to 3, restricting our outcome to three values means we cannot conclude whether our model is appropriately specified when considering the entire range of CDR scores.

Secondly, as undergraduate students with a limited understanding of medical sciences, we only analysed whether the main effects of the variables were in a particular direction rather than whether they exceeded a clinically important threshold. A more comprehensive analysis would incorporate such thresholds into the decision framework. Moreover, our loss functions were not based on scientific findings but on our own understanding of the problem, meaning the asymmetry between false positive and false negative costs was not clinically justified. Finally, diseases are usually caused by a complex interaction of many variables. Our choice to include only one interaction term (brain volume and education) and to assume linear effects of continuous predictors means other sources of misspecification may be present. Like many studies using the OASIS dataset, our results are merely generating hypothesis that would have to be validated in longitudinal studies.

Our conclusion that brain volume is associated with higher CDR scores reinforces its importance as a clinical tool for diagnosing patients and matches the widespread biological literature on this topic.

The finding that males are at a higher risk of dementia keeping age, brain volume and education could provide a new angle to the relationship between sex and dementia. However, this should be interpreted cautiously as it contradicts the general belief that women are at higher risk. This may be because our coefficients are conditioned on other variables rather than absolute effects of the sex variable. Lastly our analysis did not support the belief that higher education can help reduce the severity of dementia symptoms.

Contributions of Each Group Member

Each group member contributed roughly equally. Michael wrote the literature review, performed the sensitivity analysis for both models, and the variational inference for the complex model. Sachit wrote the problem formulation, primarily designed both models, and implemented the complex model, including the stan implementation, calibration checks, model diagnostics, evaluation of the posterior approximation, and the code correctness for both models. Sarenna set up the repository/virtual environment, processing the raw data and the exploratory data analysis, and implementing the naive model, including the stan implementation, calibration checks, model diagnostics, evaluation of the posterior approximation, and the leave-one-out for both models.

Appendix

Sources:

https://www.alz.org/alzheimers-dementia/what-is-dementia
https://www.who.int/news-room/fact-sheets/detail/dementia